

EXAM 7 – SPRING 2014

23. (2.5 points)

a. (1 point)

Discuss two reasons why the relationship between shareholders and debtholders often makes it worthwhile for a firm to minimize a specific diversifiable risk, regardless of whether the firm is an insurer.

b. (1 point)

Describe two ways the relationship between shareholders and debtholders differs for insurance firms and non-insurance firms.

c. (0.5 point)

Explain how the differences in part b. above could affect the amount of risk an insurance firm might want to retain relative to a non-insurance firm.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 23	
TOTAL POINT VALUE: 2.5	LEARNING OBJECTIVE: C6: Describe the rationale for, methods for, and effect of managing insurance and financial risks
SAMPLE ANSWERS	
Part a: 1 point	
<i>Sample Answer 1</i> (1) If shareholders (agents of debt holders) take risky gamble w/ debt holder's money, debt holders may demand higher yields & stricter bond covenants → firm should try to minimize agency issues. (2) If the firm becomes troubled, the shareholders may under invest in safer projects that provide a consistent, if lower, return ("underinvestment problem") which hurts the debt holders and the firm's viability. (The shareholders stake decreases when the firm is in distress but the bondholder's stake does not). <i>Sample Answer 2</i> Agency theory describes the conflict between shareholders and debt holders. (1) Outside of financial distress scenarios, there are conflicts between the primary goal of shareholders (optimize return on investment) and debt holders to minimize firm's default probability). As a result debt holders are risk averse. To appease debt holders, a firm may engage in risk transfer to minimize diversifiable risks that could lead to default. (2) In case of financial distress, the agency conflict heightens. Shareholders become less risk averse, as they want to take chances so that they can get some of their money out (debt holders have the first claim). Since shareholders have control of the firm's assets, this can lead to an underinvestment problem where the firm foregoes profitable investment opportunities that could benefit debt holders more. Engaging in risk reduction to reduce risk of distress helps avoid these issues.	
Part b: 1 point	
<i>Sample Answer 1</i> (1) In an insurance company the biggest "bond holders" are the policyholders. This is because the UEPR and Loss and LAE reserves dominate the balance sheet of an insurer. (2) Policyholders cannot diversify away their risk like a bondholder can by investing in multiple companies / municipalities/governments. A policyholder cannot purchase multiple policies and spread the risk. <i>Sample Answer 2</i> (1) Debt holders in insurance company can't diversify their risk (can't buy a personal accident policy from 3 insurers) → they are particularly risk averse [policyholders are also debt holders since they provide insurer liabilities of reserves & unearned premium]. Insurance Policy is a promise of future payment contingent on specific condition → policyholders who are also debtholders especially want the insurer to be stable. (2) Since the value of Liability and Unearned Premium Reserve is huge comparing to other forms of	

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capital Debt holders are especially important. If lose them, will lose both customers and source of financing.
Part c: 0.5 point
<i>Sample Answer 1</i>
An insurer may want to retain less risk since they may lose policyholders in the event of financial distress and w/ little recourse to set them back, since lowering rates will worsen distress.
<i>Sample Answer 2</i>
As policyholders, who prefer to buy from stable insurers, play an important part in insurance company capital decision making (see part b), insurance company tend to retain less risk than non-insurers.
EXAMINER'S REPORT
This question tests the understanding of the differences in the relationship between shareholders and debt holders of insurance and non-insurance firms and how they influence risk management. Most candidates were able to get at least partial credit for this question. Some candidates failed to discuss the relationship between shareholders and debt holders. Many were not able to provide sufficient reasons to the differences in the relationship.
Part a
- The candidate was expected to know that in spite of the inevitability of the conflict between shareholders and debt holders (pertaining to risk tolerance levels) there would be situations where it would be advantageous for the shareholders to adopt a risk-minimizing strategy under normal circumstances and a strategy of reducing the probability of financial distress. - The candidates needed to describe the relationships between shareholders and debt holders, describe the conflict situations, and describe the conflict resolution mechanism. - Many candidates mentioned that the interests are aligned. The divergence was not properly discussed. Many candidates made general statements instead of focusing on the relationship between shareholders and debt holders.
Part b
- The candidate was expected to know that the policyholders are the primary debt holders for an insurance company. This situation leads to a Customer/Business relationship that supersedes shareholder/debt holder relationship. Hence, in order to retain their customer base, insurance companies have to manage the risk preferences of their customers. This situation does not become an issue for non-insurance companies. - Candidates needed to mention that policyholders/customers are the primary debt holders and how insurance companies have to consider their preferences. - Most candidates identified policyholders/customers as the primary debt holders of an insurance company. Many unnecessarily brought in the issue of mutual companies; this was not part of the question.
Part c
The candidate was expected to know that the debt holder/customer would be risk-averse and would expect the insurer to remain solvent to pay claims. This emphasis would force

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insurers to reduce risk much more than non-insurance companies.

- Most candidates were able to identify customers' needs with respect to risk reduction. However, many were unable to connect this situation with risk reduction for the firm.